IMPERIAL

Stave Notation: a Diagrammatic Calculus for Specifying, Communicating, and Validating Bioimage Analysis Pipelines

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Introduction

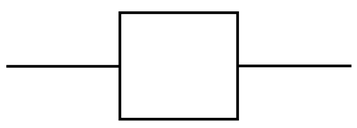
Motivation

- FAIR Principles: findability, accessibility, interoperability, reusability
- Incorporation of advanced pure and applied mathematics
- Interaction with AI advanced modern tools, e.g. Agents
- Acceleration of image analysis implementation
- Rely on visual perception of image analysis workflows

Diagrammatic Calculus

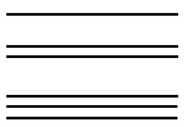
Data Flow

- Data can be represented as line(s) and morphisms/function(s) as block(s):



Data

- Lines represent where data lives:

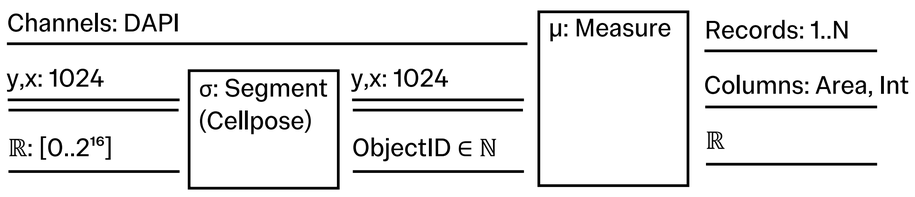


- Wire: 1D container (list, time series, table rows)
- Matrix: 2D container (spatial grid — has neighbourhood)
- Volume: 3D container

- Name above line; we can specify the cardinality of the container. Bottom line closes. Every container is labelled; the bottom line carries the value type.
- Strokes equals dimensions. ═ is literally two ── stacked (Y and X axes).

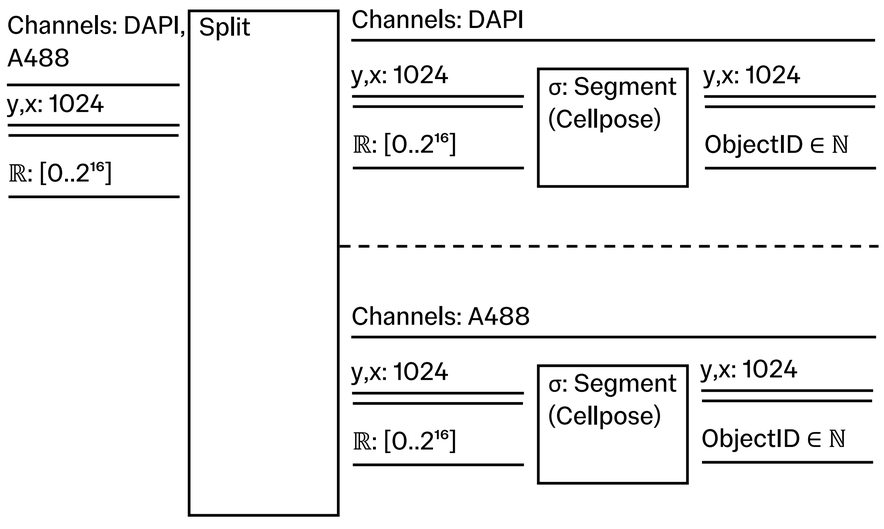
Simple Example

- Segmentation and Measurement:



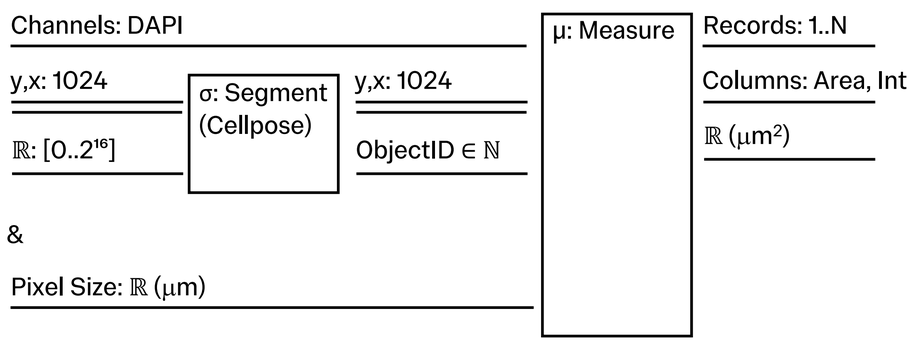
Independent Data (Data Splitting)

- Data can be split; we use dashed lines or “x” symbol:



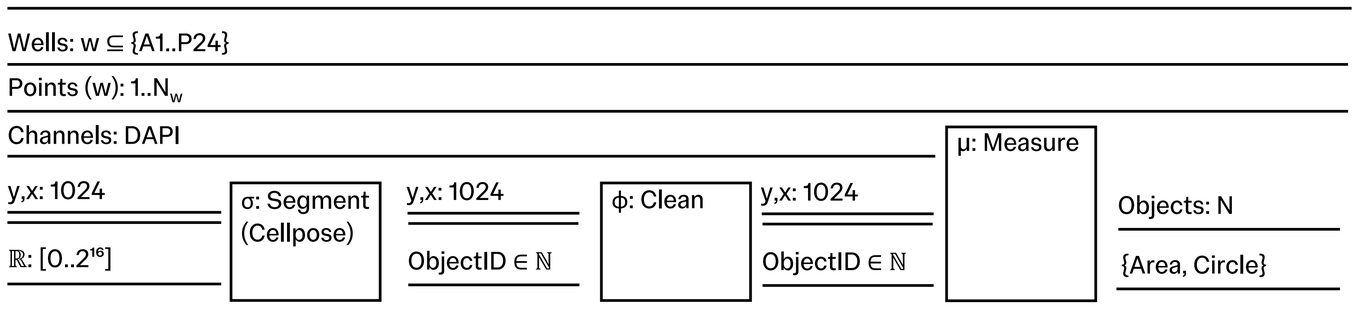
Linked Data

- Data can be linked; we use “&” symbol:



HCS Plate

- We can work with complex architecture and dependent cardinality of containers



Data Types

- Scalar:
  - ℝ — real (intensity, area, distance),
  - ℕ — natural (ObjectID, label, count),
  - ℝ: [0..2¹⁶] — ranged real (pixel value with bit depth),
  - Text — string (filename, condition).
- Coordinate:
  - (x,y,z) ∈ ℝ³ — 3D point.
- Records:
  - {Area:ℝ, Circ:ℝ, FileName:Text}.
- Reference (foreign key into another container):
  - ℕ ∈ {1..N} — index into parent,
  - ℕ³ ∈ {1..V}³ — triple of indices (mesh face → vertices).
- Geometric algebra (aspirational — constrains computation):
  - $Cl_1(3,0)$ — direction (grade-1 vector),
  - Spin(3) — orientation (rotor, forces geometric product),
  - $Cl^+(3,0,1)$ — pose (PGA motor).
- Category-enriched (names the target category, constrains morphisms):
  - (ℝ², ω) — symplectic (forces symplectomorphisms),
  - (ℝ, g) — Riemannian (forces diffeomorphisms),
  - P(ℝ) — statistical (probability, forces non-expansive maps).

Morphisms

- ι — introduce:
  - Brings data into the pipeline; load file, acquire image, read sensor,
  - Creates container lines (no input wires, only output),
  - ι: ReadND2, ι: WatchFolder, ι: Acquire.
- ϕ — filter/transform:
  - Type-preserving: ℝ → ℝ, ℕ → ℕ,
  - Box sits below container lines (Rule 3: spatial passes over).
  - fmap — applied per element, container structure unchanged,
  - ϕ: GaussianBlur, ϕ: BackgroundSubtract, ϕ: Clean.
- σ — segment:
  - Type-changing: ℝ → ℕ,
  - Intensity becomes labels,
  - Preserves spatial structure,
  - Consumes Channels if it selects one,
  - σ: Cellpose, σ: Threshold, σ: StarDist.
- μ — measure:
  - Structure-collapsing: consumes container lines,
  - Spatial (═) → flat record ({Area, Circ}),
  - Creates Objects:N by grouping pixels by ObjectID,
  - Reads & branches for calibration (PixelSize → μm²),
  - Tall box — spans multiple levels,
  - μ: ObjectMeas, μ: IntensityStats, μ: RegionProps.
- τ — restructure:
  - Changes container organisation, not values,
  - Reindex, group, sort, accumulate, join, reduce,
  - Operates at the container level it restructures,
  - τ: GroupBy, τ: Accum, τ: Join, τ: Z′-factor, τ: DoseResponse.
- ρ — change representation:
  - Same data, different mathematical structure,
  - Can change number of & branches (one in, two out or vice versa),
  - Point list → triangle mesh (creates inner &),
  - Table → image (PALM/STORM reconstruction),
  - Graph → tree or tree → graph,
  - ρ: Triangulate, ρ: Render, ρ: Skeletonise.
- ε — export:
  - Sends data out of the pipeline,
  - Save file, send notification, write to database,
  - Consumes wires (no output, or side-effect only),
  - ε: CSV, ε: Save, ε: Slack, ε: GoogleSheets.

Summary

- Mathematical precision:
  - Pipelines drawn on paper carry full data structure — containers, types, nesting
  - Mathematical foundation: cartesian double category Dbl(Img)
  - Four primitives {⊂ & × □}, seven morphisms {ι ϕ σ μ τ ρ ε}, three rules (lines are data, boxes are functions; pipeline flows left to right; the box sits at the level it operates on)
- Domain-specific language for LLMs:
  - Draw on paper → photograph → LLM generates code
  - Types constrain output: ℕ forces integers, Spin(3) forces geometric product
  - Named composites compress facility workflows into single boxes
- Arbitrarily complex data:
  - Containers nest to any depth (files → wells → fields → channels → spatial → value)
  - & (co-indexed) and × (independent) nest freely at any level
  - Recursive structures (⁝), graphs via foreign keys, meshes, point clouds, tables, flat records — all composable
- Advanced mathematics in imaging:
  - Geometric Algebra types: directions, rotors, PGA motors — typed, not just ℝ³
  - Category-enriched types (Riemannian, symplectic, statistical) constrain morphisms automatically
  - ρ converts between representations: image ↔ table ↔ mesh ↔ graph
- Contract for autonomous agents:
  - Morphism box = agent task; type signature = contract
  - Agent reads structure → selects tool, calculates parameters, validates connections
  - Multi-agent: ; = handoff, & = parallel with shared context, × = independent parallel
  - Meta-agent closes the acquisition ↔ analysis loop with intelligent microscopy
- Validated:
  - ~1,100 references, 20 fields, ~330 composites, 0 new primitives needed
  - The structural grammar of bioimage analysis is finite and closed